

MasteryLS: A Git-Native, AI-Augmented Learning Management System for Mastery-Based Learning Across Disciplines

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MasteryLS is available for evaluation at masteryls.com.

ABSTRACT

Learning management systems typically treat artificial intelligence as an add-on to a database-centric content model, and, when they support code-related assessment at all, assume a computer-science or STEM audience. We present MasteryLS, a learning management system built on two commitments: course content is stored and versioned in Git repositories rather than a proprietary database, and AI is a first-class capability across authoring, feedback, and assessment rather than a bolted-on chatbot. Instructors author Markdown content that AI can draft and refine, and every interaction, from auto-graded questions to open-ended essays and live GitHub-repository grading, is a declarative block embedded directly in that content. The system's most novel contribution is Disciplinary Reasoning Assessment (DRA), a generative simulation framework in which an instructor authors only high-level parameters and the AI generates a full professional scenario at each learner's runtime, evaluated from evidence extracted directly from the learner's own transcripts rather than a holistic AI judgment. A related Interview topic type reuses the same engine for career-readiness practice; one such run scored only 6 out of 100 with specific,

evidence-grounded critical feedback, showing the model is not tuned to be uniformly encouraging. We describe MasteryLS's architecture and report its deployment in three computer science courses at Brigham Young University, alongside prototype exploration in six additional disciplines. This is a systems and experience paper: we report design and deployment, not a controlled outcomes study, and discuss what evaluation work is needed next.

1 INTRODUCTION

Course content authoring and formative feedback remain two of the most persistent bottlenecks in higher education, and neither problem is specific to computer science. Instructors across disciplines spend disproportionate time producing and revising course material, and learners in any field benefit from prompt, substantive feedback that current staffing models struggle to provide at scale. Meanwhile, mainstream learning management systems (LMS) such as Canvas and Moodle store content in proprietary databases behind web-based editors, offering little of the version control, diffing, and multi-author collaboration that developers take for granted in Git-based workflows, and treat artificial intelligence, where it appears at all, as an external plugin rather than part of the platform's design.

We present MasteryLS, an LMS built around two commitments that run through every layer of the system. First, course content is Markdown stored in GitHub repositories, not a proprietary database: instructors get commit history, diffing, branching, and multi-author collaboration for free, and can move, audit, or fork their content independently of the platform. Second, AI is integrated as a first-class capability at every stage of the content and assessment lifecycle, namely authoring, feedback, and simulated professional practice, rather than as a bolt-on chatbot.

Although this paper is written for a SIGCSE audience and centers its deployment discussion on computer science courses, MasteryLS's content model and assessment framework are explicitly domain-agnostic. Our most novel contribution, Disciplinary Reasoning Assessment (DRA), was designed from the outset as a universal framework usable across disciplines and problem domains, and we report both real classroom deployment in computer science and prototype exploration of the same

framework in six other disciplines.

This paper makes the following contributions:

- A Git-native, domain-agnostic content model that brings software-engineering-grade version control to LMS authoring.
- An AI-assisted content-generation pipeline, integrated into the authoring workflow, that produces draft course structure and lesson content from instructor descriptions.
- A declarative interaction model in which AI grading and feedback, including rubric-graded essays, file submissions, and live GitHub-repository grading, is a first-class part of course content rather than a separate grading system.
- Disciplinary Reasoning Assessment and its Interview specialization: a generative, cross-disciplinary scenario-based assessment framework that evaluates process, competency, and disposition from evidence extracted directly from learner transcripts.
- A deployment report covering three real computer science courses and prototype cross-disciplinary exploration in six additional fields.

We are explicit about scope: this is a systems and experience paper. We have not yet collected learner-outcome or satisfaction data, and we discuss this limitation directly in Sections 7 and 8 rather than overstating what our deployment experience shows.

2 RELATED WORK

Traditional LMS platforms such as Canvas and Moodle provide mature gradebook, enrollment, and content-delivery infrastructure, but store content in platform-specific databases and offer only coarse revision history compared to a Git workflow. Git-backed, “docs-as-code” tooling has been explored specifically within computer science education: empirical work describes using Git itself as a course delivery platform [1] and documents student experiences using GitHub in software engineering courses [2]. MasteryLS extends this pattern from a supplementary tool into the LMS's canonical content store, and generalizes it beyond CS-specific repository-based assignments.

Generative AI has reshaped computing education discourse broadly [3], and a large, fast-growing body of work studies large language models (LLMs) in computer science education

specifically, including automated feedback generation, virtual teaching assistants, and code review [4]. Automated feedback for programming exercises predates the LLM era, with a systematic review cataloging over a hundred feedback-generation tools [5]. MasteryLS's github-submission interaction (Section 5.2) sits within this literature, but is designed as one interaction type among many in a discipline-agnostic authoring and grading model rather than a CS-specific autograder.

Scenario-based and simulation-based assessment has deep precedent outside CS: standardized-patient simulation has been used to appraise clinical performance in medical education since the 1960s [6], and case-based learning is a staple of business and legal pedagogy. Disciplinary Reasoning Assessment (Section 5.3) also draws on the intelligent tutoring systems tradition, where dialogue-based tutoring has been shown nearly as effective as human tutoring at raising learning outcomes [7], but generalizes both traditions: rather than authoring a fixed scenario or a fixed tutoring script per discipline, the framework generates scenarios at runtime from author-specified parameters, and evaluates six universal stages of professional reasoning that are interpreted differently by discipline.

Mastery learning [8] is the pedagogical foundation for MasteryLS's overall design: content mastery, tracked per topic, drives progression rather than a fixed calendar. Mastery learning is itself motivated by Bloom's two sigma problem [9], the finding that students who receive one-on-one tutoring outperform students in conventional group instruction by roughly two standard deviations, a level of individualized attention that is normally too costly to provide at scale. MasteryLS's AI-mediated coaching, topic-aware discussion, and teaching-interaction feedback are aimed at approximating that attention inexpensively, consistent with evidence that specific, timely feedback is one of the strongest levers on learning [10]. The teaching interaction and the DRA and Interview evaluation model in particular build on evidence that learners who explain material back to a teachable agent expend more effort and learn more, an effect sometimes called the protégé effect [11], related to the broader finding that actively retrieving knowledge, rather than passively reviewing it, improves long-term retention [12].

3 SYSTEM ARCHITECTURE

MasteryLS is a single page web application backed by three external systems, each responsible for a distinct concern. GitHub is the canonical store for course content: course structure, topic Markdown, and embedded interaction definitions are stored as ordinary version-controlled files. A managed backend stores operational data, including users, roles, enrollments, and a fine-grained progress and event log, and mediates all calls to the AI and Canvas APIs so that credentials never reach the client. An optional integration synchronizes selected topics, quizzes, and grades with a Canvas course, automatically stripping AI-only functionality, such as DRA and Interview, from that view, since those interactions have no Canvas equivalent.

This split is deliberate: GitHub content is portable, auditable, and usable independently of MasteryLS, while the backend holds only the operational state that has no meaningful existence outside a specific deployment. The same split underlies our cross-disciplinary generality claim: nothing in the architecture assumes CS-specific content, code execution, or grading.

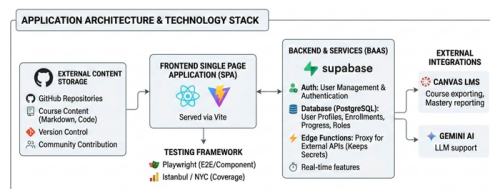


Figure 1. MasteryLS system architecture: GitHub stores canonical course content while a managed backend handles operational data and mediates AI and gradebook integrations.

4 ACCELERATING CONTENT DEVELOPMENT WITH GITHUB AND AI

Because content is Markdown in a Git repository, instructors get commit-level history, diffing, revert, and multi-author collaboration without any bespoke versioning system. An in-browser editor provides standard authoring conveniences, such as diffing against the last commit, alongside toolbar shortcuts for inserting interaction templates.

AI assistance is layered onto this workflow rather than replacing it. At course-creation time, an instructor can supply a title and description and receive an AI-generated starter structure of modules and topic stubs; at the topic level, stub topics can be expanded into first-draft lesson Markdown in batches. Finer-grained tools operate within a topic: generating a new section, rewriting only a selected range of text, running a spelling/grammar/content review pass, and generating an illustrative image. In every case the recommended and observed workflow is draft-review-refine: AI output is treated as a blueprint that an instructor curates, not a final artifact, and the underlying Git history means any AI-authored change can be diffed, reverted, or attributed like any other commit.

We do not yet have timing data comparing AI-assisted to fully manual authoring (Section 7 discusses this as a needed follow-up study). What we do have is breadth: beyond the three deployed computer science courses, the same authoring pipeline has been used, as a prototype exploration, to draft course content for Human-Computer Interaction, Electrical Engineering, Chemical Engineering, Legal, Religious Education, and Agricultural topics. None of these six was deployed to a live course with students, but the pipeline's ability to produce plausible starting material across such a heterogeneous set of domains, from a chemical-engineering process description to a religious-education lesson, is itself evidence that the authoring tools are not tuned to STEM or CS-specific content.

5 AI-DRIVEN FEEDBACK AND ASSESSMENT

5.1 Interaction Model Overview

Every gradable or interactive element in a MasteryLS topic, such as a multiple-choice question, an essay prompt, a file upload, or a live chat, is represented as a single declarative block embedded directly in the topic's Markdown next to the surrounding instructional text, rather than as a separate authoring surface or database record that must be kept in sync with the content. This lets us classify interactions along one dimension that recurs throughout the system: auto-scored objective types that require no AI at grading time; AI-graded, rubric-driven types such as essay, file-submission, url-submission, github-submission, and ai-web-page; and AI-conversational types such as teaching and prompt, in which the interaction itself is a dialogue. Two configuration options control whether a score is posted

automatically or held for manual submission, and whether the posted result includes a numeric grade or feedback only, and a mentor can always overwrite an AI-generated score or comment. Automation never removes instructor authority over the final grade.

5.2 Deep Feedback Across Interaction Types

Several interaction types are designed to produce feedback that goes well beyond a right or wrong marker. For these types the AI reads free text, a linked resource, an uploaded file, or an entire codebase, and returns qualitative, source-grounded commentary alongside a score. In every case the instructor can optionally supply a grading rubric that is not necessarily shown to the learner, and the same automatic-versus-manual scoring control described above still applies.

Essay. An essay interaction collects a free response and grades it against the essay prompt, the surrounding topic content, and, when supplied, the additional grading criteria. The returned feedback is written prose that explains what the response did well and what it missed, so a learner receives more than a percentage.

Teaching. A teaching interaction inverts the usual question and answer pattern. Rather than answering a question, the learner explains a concept to the AI across an open dialogue, and the AI evaluates the full conversation, not a single message, for correctness, depth, and how the explanation holds up under follow up questions. Feedback addresses the reasoning shown across the dialogue rather than a single submitted answer, in the spirit of recent work automatically inducing conversational tutoring scripts from lesson content [13].

Url and file submission. These two interaction types share a single underlying pattern: an artifact is acquired, the AI analyzes it, and the learner receives feedback. A url-submission interaction acquires the artifact by fetching a submitted link, optionally validating that it is reachable, while a file-submission interaction acquires the artifact directly, accepting one or more uploaded files such as a report, a dataset, or a design document. Once acquired, both are graded the same rubric-grounded way used for essays, with feedback on the artifact itself rather than on a description of it.

GitHub submission. The deepest example is github-submission. Where url-submission grades a single linked page, github-submission has the AI fetch and read the files of an entire public repository and grade them against the instructor's expected outcomes. Application code is weighted over Markdown, and large or binary files are skipped, so the model spends its attention on the parts of the repository that matter to the rubric. The result is a score from 0 to 100 and Markdown feedback that cites real file paths, telling the learner, for a rubric item such as “automated tests cover core behavior,” exactly which files were checked and where they fell short. If an instructor omits grading criteria, the submission scores 0 with an explicit “no grading criteria configured” message, so grading is never silently vague. Because this mechanism evaluates the contents of a codebase rather than a single answer, it generalizes past programming assignments to any discipline whose deliverables are file based, such as a data-analysis notebook, a document set, or a design-file archive.

Discuss and notes. Beyond graded interactions, a topic-aware Discuss panel lets a learner ask questions scoped to the current topic, or to a specific section when opened from a heading, without leaving the lesson, and useful responses can be saved as

personal notes. Discuss and the teaching interaction above are MasteryLS's most direct attempts at inexpensive one-on-one mentoring, motivated by Bloom's two sigma problem (Section 2): Discuss answers a learner's own questions on demand, while teaching has the learner explain material back to the AI, building on the protégé effect [11]. Neither feature assumes code; both operate purely over instructional Markdown and learner dialogue.

5.3 Disciplinary Reasoning Assessment (DRA)

DRA is MasteryLS's most novel contribution and its clearest evidence against a CS-specific reading of the system. Rather than authoring a fixed case study, an instructor authoring a DRA topic specifies only generation parameters: a target discipline, a problem type, a difficulty level on a scale from 1 to 5, which of two modes are enabled (Practice, Final, or both), whether random “instability” events are enabled, and the intended learning outcomes. No scenario, stakeholder, or resource is written or reviewed at authoring time. When a learner begins the assessment, the AI generates, from those parameters, a complete scenario: a narrative situation, a set of stakeholders and resources with distinct personalities and bounded knowledge, constraints, and, depending on difficulty, inflection points and standards of evidence.

The learner works the scenario through six universal disciplinary stages: Frame, Research, Model, Act, Validate, and Reflect, each interpreted differently depending on discipline (Table 1). At every point the learner can select a stakeholder or resource, converse with an AI agent playing that role, and record structured reasoning (current understanding, assumptions, unknowns, hypotheses, decisions, evidence, confidence). Additional stakeholders and resources surface automatically as they are mentioned in conversation, rather than being listed up front, so investigation itself is part of the task. In Practice mode the learner can regenerate or abandon a scenario freely and receives continuous AI coaching; in Final mode the scenario is locked after a single generation and feedback is withheld until completion.

Difficulty is not a single knob but six independently calibrated levers: how much of the scenario is disclosed up front, how specific the per-stage hints are, how cooperative stakeholders are in conversation, how directive practice-mode coaching is, how high the evidence bar is for a strong rating, and whether the generated constraints are mutually satisfiable. All six are set together so the experience shifts coherently from a scaffolded tutorial to an unguided professional challenge.

Evaluation is deliberately evidence-grounded rather than a single holistic AI judgment. An observation agent scores three dimensions: Process (the six stages), Competency (systems thinking, communication, design reasoning, evidence-based reasoning, decision-making), and Disposition (curiosity, ownership, integrity, persistence, empathy, accountability). Every attribute rating is backed by discrete evidence items extracted from the learner's transcripts, each carrying a polarity (positive or negative) and an impact weight, so a high rating with sparse or contradicted evidence cannot sustain a high score. Completed assessments accumulate into an exportable portfolio that shows a trajectory across scenarios rather than a single grade.

Stage	SE	Biology	History	Acct.
Frame	Clarify stakeholder/s/constraints	Define the biological question	Define the historical problem	Clarify the financial issue

Stage	SE	Biology	History	Acct.
Research	Gather requirements	Collect observations	Examine primary sources	Collect financial records
Model	Design architecture	Develop hypotheses/models	Construct interpretations	Build financial models
Act	Implement software	Conduct experiments	Write historical argument	Prepare statements
Validate	Test behavior	Analyze results	Evaluate evidence	Audit and reconcile
Reflect	Evaluate tradeoffs	Consider limitations	Reconsider interpretations	Assess risks

Table 1. Illustrative interpretation of DRA's six universal stages across four example disciplines, as sketched in the framework's design documentation. This table illustrates intended generality; real deployed and prototype examples follow in Figures 2 to 5 and Table 2 (Section 7).

DRA is not only a design proposal: Figure 2 shows an actual final assessment run in CS 240 (Advanced Software Construction). The generated scenario asks the learner to extend a school grading system, and the workspace view (Figure 2a) shows the learner in the Model stage, consulting an AI-played IT manager stakeholder about API design while recording architecture notes alongside a product owner and a security officer as additional stakeholders and a legacy data export and a specification draft as resources. Figure 2b shows the resulting evaluation for that same run: an overall score of 92 out of 100, rated Exemplary, computed from a Process score of 100, a Competency score of 96, and a Disposition score of 70, with 42 supporting evidence items such as “identified the 128MB RAM limit for the CLI and estimated the JVM footprint to ensure compliance,” each contributing a small, explicit support weight to its attribute score rather than a single opaque AI judgment. This is one run, not a systematic sample, but it is real deployed evidence rather than a hypothetical illustration.

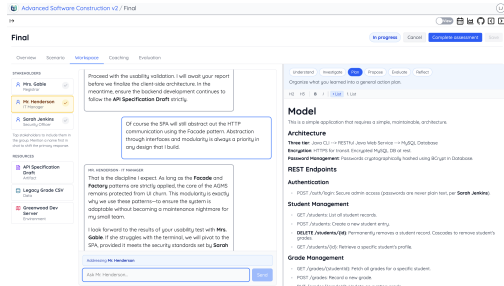


Figure 2a. DRA Workspace view from a CS 240 final assessment: the learner interviews an AI-played IT manager stakeholder while recording notes in the Model stage.

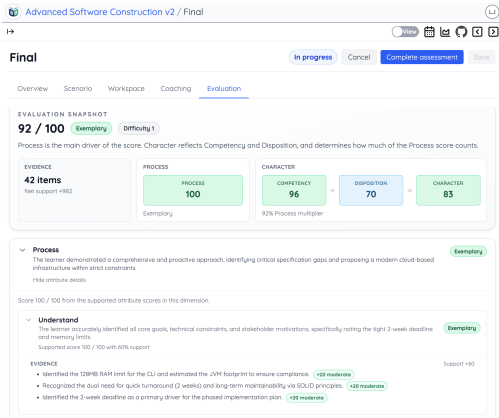


Figure 2b. DRA Evaluation view for the same run, showing the Process, Competency, and Disposition scores and expandable, evidence-backed attribute detail.

DRA has also been prototyped, outside any live course, in non-CS settings. Figure 3 shows a “Christian Outreach” scenario built for ministry and congregation-strengthening training: the stakeholders are a missionary companion, a local lay leader, a mission president, and a person the learner is teaching, and the Investigate stage has the learner reviewing outreach resources and planning who to visit and why. A separate prototype scenario, not shown here, places the learner on a startup's board of directors and asks them to recommend a survival plan from a shrinking cash runway, giving early evidence that the same six-stage engine holds up for business strategy reasoning as well as for the technical and ministry scenarios already illustrated.

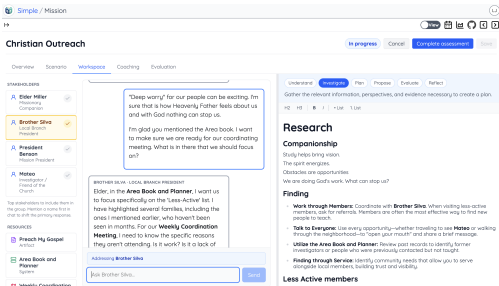


Figure 3. DRA Workspace view from a prototype “Christian Outreach” scenario built for Religious Education: the learner consults a local lay leader while planning ministry visits in the Investigate stage.

5.4 Interview: An Applied Specialization

Interview reuses DRA's generative-scenario and evidence-scored evaluation engine for a narrower, more immediately legible use case: job-interview practice. An instructor specifies a discipline, job title, job description, difficulty, learning outcomes, and enabled modes; at runtime the AI generates a multi-session interview with one or more AI-played interviewer personas, and the learner practices across sessions, optionally requesting coaching between them. Evaluation reuses DRA's Competency and Disposition scoring machinery alongside a per-session rating, combined into a single overall score and rating band using the same evidence-supported calculation. Interview matters to this paper's argument less for its own sake than for what it demonstrates: the DRA evaluation engine is reusable across a structurally different assessment shape, not hand-built for one scenario type, which is independent evidence for the generality of the underlying framework.

Figure 4 shows a prototype Interview built for an Electrical Engineering hiring context: an energy storage company interviewing candidates for a senior grid-resiliency engineering role across four AI-played interviewers spanning recruiting, power-systems engineering, project management, and engineering leadership (Figure 4a). Unlike the CS 240 DRA run in Figure 2, this run scored only 6 out of 100, rated Beginning, with specific critical feedback flagging missing knowledge of load-flow studies and utility safety codes, immature responses to a senior-level question, and impractical troubleshooting suggestions (Figure 4b). We include this low-scoring run deliberately: it shows the evidence-grounded evaluation model is not tuned to be uniformly encouraging, and can produce specific, critical feedback grounded in named evidence when performance warrants it.

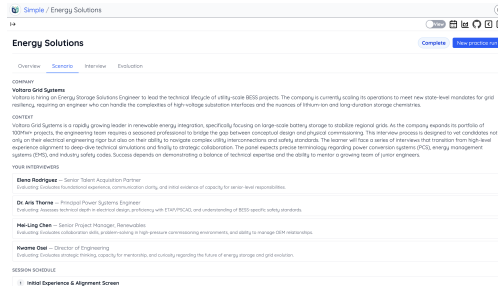


Figure 4a. Interview scenario view for a prototype Electrical Engineering role: an energy storage company interviewing for a grid-resiliency engineering position.

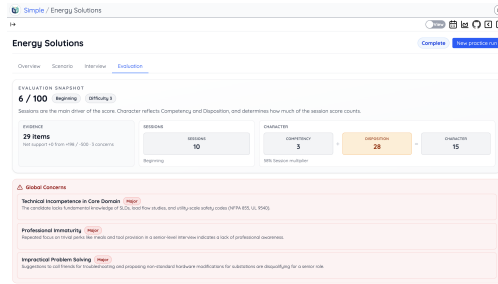


Figure 4b. Interview Evaluation view for the same run: a score of 6 out of 100 (Beginning) with named, evidence-grounded concerns.

6 MASTERY TRACKING AND ANALYTICS

Three learner- and instructor-facing views close the loop on the interaction and content model described above. MasteryView summarizes course-level mastery, including completion percentage, time spent, and interaction coverage, for a single learner or, for editors, across an entire class. Progress is a filterable, paginated event log of every recorded action usable for audit or self-review. Metrics aggregates time-on-task and activity trends across courses, learners, and date ranges. An Observe mode lets an editor view a course exactly as a specific learner would, read-only, for support and debugging. These views are also the intended data source for the outcome study we propose as future work (Section 9): the event log already records the granularity a learning-outcomes analysis would require.

7 DEPLOYMENT EXPERIENCE ACROSS DISCIPLINES

We report deployment experience, not a controlled evaluation: we have not yet collected learner-outcome measures or a student- or instructor-satisfaction survey from any MasteryLS course, and we

say so explicitly rather than implying rigor we do not have. Deployment experience falls into two tiers that we keep clearly distinct throughout this section.

Real deployment. MasteryLS has been deployed in three live computer science courses at Brigham Young University: CS 260 (Web Programming), CS 240 (Advanced Software Construction), and CS 329 (Quality Assurance and DevOps).

CS 240 has used DRA as a final assessment, illustrated in Figure 2. CS 260 and CS 329 have both used standard interactions, AI content generation, and github-submission grading, together with Interview.

Prototype exploration. Independent of live deployment, we used MasteryLS's content-generation and DRA/Interview pipelines to build course material and disciplinary scenarios for six additional fields, namely Human-Computer Interaction, Electrical Engineering, Chemical Engineering, Legal, Religious Education, and Agricultural topics, specifically to test whether the tooling holds up outside computer science. None of these six has been deployed to a live course with students; their value is as an existence proof of cross-domain feasibility, not as deployment or outcome evidence. Table 2 summarizes both tiers.

Discipline	Tier	Content	DRA/Int.	Std. Int.
CS 260 (Web Prog.)	Deployed	Yes	Yes	Yes
CS 240 (Adv. SW Constr.)	Deployed	Yes	Yes	Yes
CS 329 (QA/DevOps)	Deployed	Yes	Yes	Yes
HCI	Prototype	Yes	—	—
Electrical Eng.	Prototype	Yes	Yes	—
Chemical Eng.	Prototype	Yes	—	—
Legal	Prototype	—	Yes	—
Religious Ed.	Prototype	Yes	Yes	—
Agricultural	Prototype	Yes	—	—

Table 2. Discipline $\times$ MasteryLS capability, split by deployment tier. A dash (“—”) indicates the capability has not been used or attempted for that combination.

Figure 5 illustrates the content-generation side of that prototype exploration directly, distinct from DRA/Interview: an AI-generated Agricultural Education topic explaining ruminant digestion, written in an engaging voice with a worked four-stage breakdown and an accompanying generated diagram, produced from the authoring pipeline described in Section 4 rather than hand-written by an instructor.

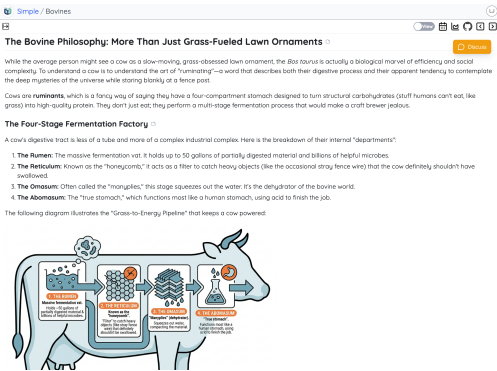


Figure 5. AI-generated topic content for an Agricultural Education prototype: explanatory text and an accompanying generated diagram covering ruminant digestion.

We see this two-tier picture as a reasonable, honestly reported first step: three real deployments establish that the system works

in production for a real course, and six cross-domain prototypes establish that nothing in the design is CS-specific. Neither tier yet establishes that learners in any discipline learn more, faster, or better because of it, which is the natural next study (Section 9).

8 DISCUSSION

MasteryLS's Git-native content model offers benefits independent of AI: content is portable across platforms, auditable through ordinary Git tooling, and open to multi-author collaboration without a bespoke revision system. Layering AI onto that model, rather than into a separate database-backed authoring surface, keeps AI-authored and AI-graded content subject to the same versioning, diffing, and revert affordances as everything else an instructor writes. The DRA framework is, we think, the paper's strongest evidence that this generality is not merely architectural: real deployment in computer science and structured prototype exploration in six other fields, spanning both technical (Electrical Engineering, Chemical Engineering) and non-technical (Legal, Religious Education) content, is a more convincing existence proof than a design argument alone.

We are equally direct about what that evidence does not show. Deployment in three courses at one institution, and prototype exploration with no live students in six more fields, tells us the system runs and produces plausible content and scenarios; it does not tell us whether learners master material faster, whether AI-generated feedback is as reliable or fair as instructor feedback, or whether generated DRA scenarios hold up to sustained, discipline-expert scrutiny outside CS. AI grading reliability and bias, academic-integrity questions raised by AI-assisted authoring and AI-assisted submissions [14], and the security posture of credential handling for GitHub and third-party AI services are all open risks that a systems paper of this scope can describe but not resolve. Instructor-facing controls, namely score override, explicit and inspectable grading criteria, and the choice between automatic and manual grade posting, are the mechanisms currently in place to mitigate, though not eliminate, these risks, and we view them as a starting point rather than a complete answer.

9 CONCLUSION AND FUTURE WORK

We presented MasteryLS, a Git-native, AI-augmented LMS whose content model, authoring pipeline, and assessment framework are designed to generalize beyond computer science, and reported its deployment in three real BYU CS courses alongside prototype exploration in six additional disciplines. The paper's central artifact is Disciplinary Reasoning Assessment: a generative, evidence-scored simulation framework for authentic professional reasoning assessment, together with its Interview specialization for career-readiness practice. Our priority for future work is closing the evaluation gap this paper is explicit about: a learner-outcome and satisfaction study drawing on the existing CS deployments and, ideally, a first real non-CS deployment converting one of our six prototype disciplines into a live course; and a timed or comparative study of AI-assisted versus manual content authoring. Longer term, we intend to explore adaptive sequencing and richer cross-discipline analytics built on the progress data the system already records.

ACKNOWLEDGMENTS AND AI-USE DISCLOSURE

Generative AI (Anthropic Claude) was used to assist with drafting and organizing this manuscript from the author's design documentation and prior notes. All technical claims were reviewed by the author, who takes full responsibility for the content, consistent with ACM's policy on authorship and generative-AI use.

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